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EDUCATION

University of California, Davis

Fall 2026 - Present

Masters of Science, Electrical and Computer Engineering

University of California, Davis

Fall 2021 - Winter 2026

Bachelor of Science, Electrical Engineering

GPA: 3.8

• **Associations:** Tau Beta Pi (Honors Engineering Society), E-SEARCH

• **Awards:** Dean's Honor List: 4X, Top 3 in competition for Robotics

PROJECTS

Decoding Neurological Disorders

Researcher, Fall 2023

Advisor: Kevin Immanuel Gubbi (Ph.D. Candidate)

- Focus Area: Investigating calcium disruption in neurological disorders through hardware Trojan modeling and simulation.
- Key Project: Simulated Alzheimer's Disease to leverage hardware advancements for rapid treatment discovery.
- Tools/Technologies Used: Verilog, SystemVerilog.
- Highlights:
 - Developed simulations that proactively address malicious hardware vulnerabilities.
 - Explored potential for hardware-based solutions to function as medicinal aids.
 - Enhanced clinical testing efficiency while eliminating risks to patients.
 - Contributed to refining on-chip learning systems to emulate a healthy brain model.

Engineering Projects

8-bit Barrel Shifter (Winter 2026)

- **Objective:** Design an 8-bit barrel shifter from transistor level
- Progressed from Inverter → Combinational blocks → 2 to 1 MUX → 8-bit barrel shifter, developing schematics and physical layouts for each stage.
- Simulated schematics to verify expected functionality
- Ran DRC and LVS verification, confirming design-rule compliance and schematic-to-layout accuracy

Autonomous Camera-based Vehicle (Winter-Spring 2025)

- **Objective:** Design and build a self-driving vehicle capable of following a track using only data given by its camera
- Hardware: Developed prototype hardware design on Altium for autonomous vehicle, including motor control via H-Bridge drivers, battery-relative voltage control via voltage regulators, MOSFETs and passive components for noise filtering, and integration for OpenMV microcontroller and servo
- Software: used Python that analyzes frame-by-frame data using line-detection algorithms and calculating error/deflection angle for directional decision making and adjusting PWM accordingly. Used PID controller to control speed and direction correction.

Biometric Attendance (Winter 2025)

- **Objective:** Scan a student's finger print to record attendance and store data on cloud, accessible through spreadsheet
- Hardware: Integrated a generic fingerprint scanner module with UART communication via CC3200, LCD display for user interface, IR-Receiver and AT&T remote to navigate through user interface, RC noise filtering for IR-Receiver
- Software: Implemented an embedded system using C. Converted Arduino library for Adafruit fingerprint sensor to be compatible with C language. Translates remote button press into either directional keys to navigate UI or alphanumeric symbols to store student name + ID number with fingerprint.
- Cloud Storage: utilized AWS Cloud and Lambda function to store real-time attendance into a downloadable spreadsheet

Sound-Controlled Movement Robot, Top 3 Finalist | University Open House Presenter (Winter 2023)

- **Objective:** Integrating analog circuits, digital systems, and signal processing to create a motor-driven robot that navigates the fastest path toward a sound source.
- Implemented key features including:
 - Signal detection using dual microphones and op-amp amplifiers.

- Signal digitization and processing via TI MSP432 microcontroller.
 - Control systems utilizing GPIOs, ADCs, timers, periodic interrupts, and PWM.
- Achieved a **Top 3 placement** in the course competition and showcased the robot at the university open house, advising prospective students interested in electrical engineering projects.

AI Image Classifier: Art Period Recognition (Spring 2022)

- **Objective:** Designed and developed a MATLAB application to classify images based on art periods using AI-driven algorithms.
- **Highlights:**
 - Implemented machine learning techniques to analyze and categorize artistic styles from various periods.
 - Enhanced algorithm accuracy through feature extraction and data preprocessing. (~70-90% Accuracy)
 - Utilized MATLAB for application development and visualization of classification results.
 - Showcased technical skills in AI, image processing, and programming.

OTHER EXPERIENCE

Tutor | *Tau Beta Pi, Davis, California*

Spring 2024 – Current

- Tutored following classes: Calculus I-III, Vector Calculus, Physics (Classical Mechanics, Electromagnetism), Circuit Theory

Key Contributions

- Mentored a student from a D to a B in Calculus III through targeted instruction and personalized study strategies.
- Guided a student at risk of university dismissal to successfully pass his Circuit Theory examination, resulting in his successful readmission to the university and academic recovery.
 - Initially provided tutoring through Tau Beta Pi, and due to exceptional satisfaction and results, transitioned to private, low-cost tutoring for more personalized support
- Empowered students to develop problem-solving skills and confidence in tackling complex mathematical and engineering concepts.
- Recognized by many students for fostering a supportive learning environment and achieving measurable improvements in student performance.

SKILLS

Technical Skills: FPGA Development, Verilog, System Verilog, Intel Quartus, RISC-V, MATLAB, C/C++, Python, Excel, Engineering lab equipment, Cadence, Silvaco/Athena, Semiconductor fabrication in a cleanroom environment